Unrelated Is Not Opposed: A Signed Scale for Meaning Preservation

Anonymous ACL submission

Abstract

Meaning preservation metrics output a non-negative score and cannot represent opposition: MeaningBERT scores "I drink milk" and "I do not drink milk" at 58.30 out of 100. We propose a signed scale on $[-100, 100]$ that weights an existing similarity metric by the probability, given by a polarity head, that one sentence contradicts the other, so unrelated sentences score 0 and contradictions score below it. A polarity head trained on inference data with generated identical, unrelated, opposite and antonym pairs catches negations and most unseen antonyms, which a negation word list cannot. With an unchanged metric, the signed score accepts eleven times fewer contradictions as meaning-preserving at a fixed threshold.

1 Introduction

Assessing whether a rewritten sentence preserves the meaning of its source is a prerequisite for evaluating text simplification, summarization and translation (Saggion, 2017). The metrics built for this purpose, whether embedding-based (e.g. Sentence-BERT (Reimers and Gurevych, 2019) or BERTScore (Zhang et al., 2020)), question-based (e.g. QuestEval (Scialom et al., 2021)) or trained end to end (e.g. LENS (Maddela et al., 2023) or MeaningBERT (Beauchemin et al., 2023)), return one non-negative number, from 100 for identical meanings to 0 for unrelated sentences.

This output cannot represent opposition. Two contradictory sentences are about the same topic and share most of their words, and a metric of shared meaning reads that overlap as preserved meaning. MeaningBERT scores "I drink milk" and "I do not drink milk" at 58.30 out of 100, in the upper half of the scale. Following de Marneffe et al. (2008), we call two sentences contradictory when they are extremely unlikely to be true of the same situation. This covers explicit negations such as this pair, and also contrary events such as "a girl is running outside" and "a girl is running inside", which SICK labels as contradictions because both captions describe the same scene. Baral (2026) observed the same effect on legal text, where edits that reverse the legal force of a sentence barely move embedding metrics.

We therefore propose a signed scale on $[-100, 100]$, on which 100 denotes identical meanings, 0 unrelated sentences, and negative values related sentences that assert opposite facts. We obtain it by composing the score of a similarity metric with a separately trained polarity head. Our contributions are:

1. A meaning preservation metric on a signed scale that distinguishes opposed meaning from unrelated meaning (Sections 3 and 7).
2. A polarity head, the stratified, leakage-free corpus with derived-polarity data augmentation on which it is trained, and a lexical augmentation that teaches it antonymy (Sections 4 and 7).

2 Related Work

2.1 Meaning Preservation Metrics

Meaning preservation metrics fall into three families. Embedding-based metrics compare sentence representations: Sentence-BERT computes the cosine between two sentence embeddings, and BERTScore matches each token to its most similar counterpart in the other sentence. A negation adds as little as one token, which barely changes either score. Question-based metrics such as QuestEval generate questions from one text and check whether the other answers them, and a negated sentence, which mentions the same entities, answers most of them. Learned metrics such as LENS and MeaningBERT replicate human ratings, of overall simplification quality for the former and of meaning preservation for the latter. These ratings run from identical meaning down to no shared meaning,

with no value for opposite meaning, so an annotator can only rate a contradiction low, at the end reserved for unrelated pairs. The training signal cannot separate the two, and in practice the learned metric does not even score contradictions low, as the milk pair shows.

All these metrics are validated by correlation with human ratings, and MeaningBERT adds two sanity checks, under which identical sentences must score 100 and unrelated sentences 0. None tests opposition: any monotone function of lexical overlap passes both checks (Baral, 2026), and ratings that do not distinguish a contradiction from a poor paraphrase cannot reward a metric that does. Specialized metrics for legal text (Beauchemin et al., 2025) and biomedical simplification (Lyu and Pergola, 2024) show that general metrics miss small edits with large consequences, but they too output a non-negative score. We keep such a metric as the similarity component of a signed score and supply the missing direction separately (Section 3).

2.2 Natural Language Inference

Natural language inference provides the polarity that preservation metrics lack. Large corpora (Williams et al., 2018; Nie et al., 2020) and fact-verification datasets (Thorne et al., 2018; Schuster et al., 2021) label sentence pairs as entailment, neutral or contradiction, and suites such as MoNLI (Geiger et al., 2020) and NaN-NLI (Truong et al., 2022) probe negation. The two tasks are complementary: an inference model says whether the second sentence follows from or denies the first but not how much meaning they share, and a preservation metric measures the shared meaning but not its direction. We therefore combine them (Section 3), and SICK (Marelli et al., 2014), to our knowledge the only resource annotating both over the same pairs, serves to evaluate the combination. Inference models already serve evaluation: SummaC (Laban et al., 2022) and AlignScore (Zha et al., 2023) detect inconsistency between a summary and its source, and MENLI (Chen and Eger, 2023) combines inference scores with similarity metrics to resist adversarial edits such as negation. These metrics still output a non-negative score, whereas ours keeps the similarity metric and makes the score signed, so an unrelated pair stays at 0 and a contradiction falls below it.

3 A Signed Scale

Meaning metrics use an unsigned scale on which 100 denotes identical meaning and 0 unrelated sentences, the two anchors of the sanity checks of Beauchemin et al. (2023). We propose a signed scale on $[-100, 100]$. Its positive half is the unsigned scale with both anchors unchanged. Its negative half denotes opposite meaning: the sign carries the polarity and the absolute value the closeness of the two sentences, so -100 denotes a pair as close as possible in content that asserts the opposite. We compose the two quantities as

$$s = m \times (1 - 2\,p_{\text{contra}}), \tag{1}$$

with $m \in [0, 100]$ the score of any similarity metric and $p_{\text{contra}} \in [0, 1]$ the probability of contradiction given by a polarity head. The polarity head is a classifier that reads the two sentences and, as in natural language inference, outputs a probability for entailment, neutral and contradiction. Only p_{contra} enters Equation 1, and Sections 4 and 5 describe how we train the head. The form is a product and not a difference, so an unrelated pair, with $m \approx 0$, scores near zero whatever its polarity, whereas a subtraction would let it drift negative. Figure 1 places five pairs built on "I drink milk" on the scale.

One may object that a negation is closer in meaning to its source than an unrelated sentence, so that 100, 50 and 0 would be a defensible order for a paraphrase, a contradiction and an unrelated pair. This order conflates contradictions with mildly related pairs, since a contradiction and a paraphrase that keeps half of its content both score near 50. A signed scale keeps the closeness in $|s|$ and adds the direction. Human raters confirm that closeness: on the 1–5 relatedness scale of SICK, they rate contradictions 3.59 on average and neutral pairs 2.98, in line with the observation that opposites are semantically close (Mohammad et al., 2013; Hill et al., 2015). Placing contradictions below zero also reflects their cost when a metric decides whether a rewrite may replace its source. An unrelated output is visible to the reader, but a negation misleads, and factual errors are a central risk of rewriting (Maynez et al., 2020), to the point that an inaccurate simplification may be worse than none (Devaraj et al., 2022).

The factor 2 makes the scale symmetric and follows from the definition, not from a fit. Since $1 - 2\,p_{\text{contra}} = (1 - p_{\text{contra}}) - p_{\text{contra}}$, m is

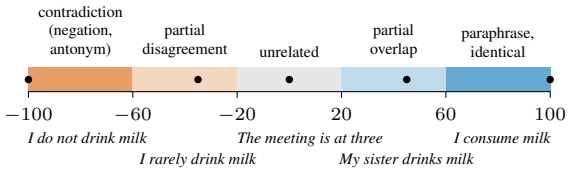


Figure 1: The signed scale, with indicative zones. Dots: pairs formed with "I drink milk".

weighted by the margin between the probabilities that the pair does not contradict and that it does. A certain non-contradiction leaves m unchanged and a certain contradiction maps it to $-m$, so only the sign changes and s stays within $[-100, 100]$ with no clipping. The sign flips at $p_{\text{contra}} = 0.5$, and a pair the head leans towards contradiction lands in the negative half in proportion to its closeness, as "I rarely drink milk" does in Figure 1. A factor of 1.5 would truncate the negative half at $-m/2$, and a larger factor would flip the sign on weaker evidence and saturate at -100 (Section 7). The two ends are anchors: a paraphrase that keeps all the meaning scores $+100$, and the exact negation, identical in content but asserting the opposite, scores -100. In practice, a preservation metric gives a negation a lower m than its closeness warrants, and MeaningBERT would place the milk negation at -58.30.

4 Data

4.1 What Each Corpus Contributes

The signed score needs three things from the data: many inference pairs to train the polarity head, a corpus annotated with both relatedness and inference to study how m and p_{contra} combine, and targeted sets that expose the failures an aggregate score hides. No single corpus provides all three, so we draw from the five corpora of Table 1.

VitaminC (Schuster et al., 2021) supplies the volume, in the regime this paper targets. Its 488,904 pairs are contrastive by construction: a Wikipedia sentence was edited so that the claim supported by its previous revision is refuted by the next, which yields contradictions that differ from their source by a few tokens and not by topic. SICK (Marelli et al., 2014) links the two quantities, with a relatedness score and an inference label on the same pairs. MoNLI (Geiger et al., 2020) and NaN-NLI (Truong et al., 2022) probe negation. MoNLI has 1,202 pairs that differ by one word replaced with a more general or more specific one under a negation that reverses the entailment: "There is a man not wearing a hat" entails "There is a man not wearing a sunhat", and without the negation the entailment runs the other way. NaN-NLI has 258 pairs that target negation below the clause level, scoped over a quantifier or a phrase, as in "Not all people have had the opportunities you have had" and "Some people have not had the opportunities you have had". Both never enter training and serve as held-out sets (Table 2), since a head that scores well on VitaminC while failing them has learned the corpus and not the relation. Last, CSMD (Beauchemin et al., 2023), the corpus of human meaning preservation ratings on which MeaningBERT was trained, supplies 407 test pairs to check the signed scale on simplification output.

Corpus	Rows	Labels	Use
VitaminC	488,904	FACT3	train/dev/test
SICK	9,840	NLI3 + rel.	train/dev/test
MoNLI	1,202	ent./neutral	held-out
NaN-NLI	258	NLI3	held-out
CSMD	1,355	rating 0–100	scale check

Table 1: Corpora used, with their size, label scheme and role.

4.2 Data Augmentation at the Ends of the Scale

Inference corpora contain few identical and unrelated pairs, since neither poses an interesting inference problem. Yet a signed score must be exact on them: 100 for a sentence paired with itself and 0 for two sentences with nothing in common. We therefore train under two conditions, RAW, the merged corpus alone, and AUG, the same corpus plus 25,000 generated pairs of each of three kinds. An identical pair, (A, A), is labeled entailment, so a sentence paired with itself scores 100. An unrelated pair joins two random training sentences that share few tokens and is labeled neutral, so unrelated sentences score 0 and not a negative value. A mirrored contradiction swaps an existing contradiction, (B, A) for (A, B), and keeps its label, so a contradiction is detected in either order.

A generated pair receives a label only when that label follows from the original pair. A contradiction survives the swap, since if A contradicts B then B contradicts A. An entailment does not: "a dog is running" entails "an animal is running" but not the reverse, so we mirror contradictions only. Mirroring all of them would double that class, from 50,267 to 100,534 pairs against 50,602 entailments, and push the head to overfit on contradictions. We therefore mirror a random sample of 25,000, the quota of the other two kinds, so each class grows

Split	Ent.	Neutral	Contra.	Total
Training, differs by condition				
RAW	50,602	51,191	50,267	152,060
AUG	75,602	76,191	75,267	227,060
Evaluation, identical for RAW *and* AUG				
Dev	3,939	3,864	3,968	11,771
Test	4,000	4,000	4,000	12,000
Sanity	1,500	1,500	1,500	4,500
Held-out, never trained on				
MoNLI	601	601	–	1,202
NaN-NLI	97	44	117	258

Table 2: Final polarity data per split and class. AUG adds 25,000 identical pairs, unrelated pairs and mirrored contradictions to RAW. We use the same splits for the dev, test and sanity suites.

by 25,000 and the three stay balanced (Table 2).

4.3 Stratified Splits

The development and test splits are stratified by class and by corpus. Each holds 4,000 pairs per class, drawn equally from each corpus, and a corpus too small for its share is taken whole. Without this, VitaminC, fifty times larger than SICK, would leave SICK at 8.16% of the test split and model selection would rest on Wikipedia claims alone, whereas with it SICK makes up 34.30% of the test split. Since VitaminC reuses one evidence sentence across many claims, we then remove from the development and training splits every pair whose source sentence appears in a later split. Table 2 gives the counts per split and class, and Appendix A the construction.

5 Experimental Setup

Encoders. We fine-tune the polarity head on seven pretrained encoders (Table 3), all of which read the two sentences jointly, the standard architecture for sentence-pair classification. They vary along two axes. The first is the pretraining generation, from BERT (Devlin et al., 2019) to RoBERTa (Liu et al., 2019) and DeBERTaV3 (He et al., 2023): `BERT-base` is the encoder of MeaningBERT, and DeBERTaV3 outperforms the other two on inference benchmarks at equal size (He et al., 2023). The second is the task a checkpoint was already fine-tuned on: none, semantic similarity for $\texttt{RoBERTa-base}_{\texttt{STS-B}}$, or inference for $\texttt{RoBERTa-large}_{\texttt{MNLI}}$ and the two $\texttt{DeBERTa-v3}_{\texttt{NLI}}$ checkpoints. An inference checkpoint already has a head trained on our three classes and is what a practitioner would use off the shelf (Appendix D), so including one tests whether our corpus still adds anything, and pairing each DeBERTaV3 size with its plain checkpoint isolates the effect of inference pretraining. Checkpoints that have a three-way inference head keep it, with its rows reordered when the published class order differs from ours.

Training. Each encoder is trained under RAW, with 152,060 pairs, and AUG, with 227,060, over ten random seeds, with the settings and hardware of Appendix C, and $\texttt{DeBERTa-v3-large}_{\texttt{NLI}}$ also under the two lexical conditions of Section 7.

Evaluation. We report four measures: macro-F_1 (Sokolova and Lapalme, 2009) on the test split of Table 2 for the three-way task, the area under the ROC curve (AUC; Hanley and McNeil, 1982) of $1 - p_{\text{contra}}$ at ranking entailment above contradiction, accuracy on three sanity suites, and accuracy on MoNLI and NaN-NLI. The AUC measures the quantity the signed score consumes, with 50% as chance, and for a similarity metric it ranks the score directly (Appendix B). The sanity suites hold identical, unrelated and mirrored pairs generated as in Section 4 from test-split sentences and check the ends of the scale.

Baselines. Three baselines without a pretrained encoder bound the polarity head from below. `Majority` always predicts the same class. `Jaccard` computes the Jaccard overlap (Jaccard, 1912) of the two word sets and predicts neutral below a first threshold, contradiction between the two thresholds and entailment above the second, both fitted on the development split. `TF-IDF` trains a logistic regression on word unigrams and bigrams. For the signed scale, a fourth baseline, the `Negation lookup`, flips the sign of m when exactly one sentence contains a negation word (*no*, *not*, *n't*, *never*, *nobody*, *none*, *nothing*, *nowhere*, *neither*, *nor* or *without*).

Statistics. We compare RAW and AUG for each encoder with Welch's t-test (Welch, 1947) over the ten random seeds, and two encoders with a paired t-test matched by random seed. Each p-value comes with Cohen's d, since with standard deviations near 0.20 points a difference of 0.10 points can be significant yet negligible. As we compare RAW and AUG fourteen times, we also apply the Holm-Bonferroni correction (Holm, 1979).

6 Results

The grid asks whether the polarity the signed scale needs is learnable, what data augmentation

Encoder	Condition	Macro-F_1	Sanity suites	NaN-NLI	MoNLI
DeBERTa-v3-large$_{\text{NLI}}$	RAW	**91.10 ± 0.14**	90.35 ± 0.47	**72.95 ± 0.65**	**34.93 ± 3.13**
	AUG	90.98 ± 0.31	98.76 ± 0.04	71.90 ± 1.83	31.23 ± 5.11
	LEX	91.04 ± 0.12	90.47 ± 0.46	72.67 ± 0.67	17.74 ± 3.19
	AUG+LEX	90.89 ± 0.14	**98.96 ± 0.10**	72.29 ± 1.26	19.20 ± 2.56
DeBERTa-v3-large	RAW	90.91 ± 0.10	88.89 ± 0.42	70.62 ± 1.17	22.79 ± 5.22
	AUG	90.70 ± 0.43	98.70 ± 0.40	69.11 ± 1.24	15.97 ± 5.84
RoBERTa-large$_{\text{MNLI}}$	RAW	89.52 ± 0.10	84.59 ± 1.07	68.06 ± 1.02	7.02 ± 1.10
	AUG	89.70 ± 0.10	98.69 ± 0.07	66.59 ± 1.09	8.34 ± 1.68
DeBERTa-v3-base$_{\text{NLI}}$	RAW	89.48 ± 0.15	87.46 ± 0.37	68.57 ± 0.62	11.47 ± 2.39
	AUG	89.56 ± 0.17	98.74 ± 0.08	67.40 ± 0.67	11.62 ± 2.05
DeBERTa-v3-base	RAW	88.81 ± 0.63	86.32 ± 1.74	66.40 ± 1.68	13.42 ± 2.96
	AUG	88.73 ± 0.35	98.60 ± 0.13	64.11 ± 1.92	10.87 ± 3.27
RoBERTa-base$_{\text{STS-B}}$	RAW	85.91 ± 0.15	78.67 ± 0.34	63.76 ± 0.99	4.60 ± 0.60
	AUG	86.22 ± 0.13	98.65 ± 0.08	62.60 ± 0.74	4.94 ± 0.64
BERT-base	RAW	77.72 ± 0.38	79.38 ± 0.65	46.36 ± 2.70	13.64 ± 5.52
	AUG	78.45 ± 0.26	98.31 ± 0.10	49.03 ± 1.13	17.48 ± 2.57

Table 3: Polarity head per encoder and condition, mean ± standard deviation over ten random seeds, in percent. **Bold**: best per column.

Model	Preserv. r	Polarity AUC	Polarity Macro-F_1
Similarity metrics			
MeaningBERT	0.32	60.81	–
BERT-base	0.64	53.73	–
DeBERTa-v3-large	**0.71**	81.51	–
Polarity heads, off-the-shelf then ours			
DeBERTa-v3-large$_{\text{MNLI}}$	–	96.62	68.36
DeBERTa-v3-large$_{\text{NLI}}$	–	**98.41**	**91.10**
Baselines			
Majority	–	–	16.67
Jaccard	–	63.49	39.36
TF-IDF	–	68.53	45.64

Table 4: Preservation (r with human ratings on 1,536 pairs of a merged preservation test set) and polarity (12,000 test pairs).

Encoder	Macro-F_1 Δ (pp)	Macro-F_1 d	Sanity suites Δ (pp)	Sanity suites d
DeBERTa-v3-large$_{\text{NLI}}$	-0.11$_{\pm 0.11}$†	-0.47	+8.40$_{\pm 0.15}$	+25.43
DeBERTa-v3-large	-0.21$_{\pm 0.14}$†	-0.67	+9.81$_{\pm 0.18}$	+24.07
RoBERTa-large$_{\text{MNLI}}$	+0.19$_{\pm 0.05}$	+1.79	+14.10$_{\pm 0.34}$	+18.57
DeBERTa-v3-base$_{\text{NLI}}$	+0.08$_{\pm 0.07}$†	+0.49	+11.28$_{\pm 0.12}$	+42.24
DeBERTa-v3-base	-0.08$_{\pm 0.23}$†	-0.15	+12.28$_{\pm 0.55}$	+9.94
RoBERTa-base$_{\text{STS-B}}$	+0.31$_{\pm 0.06}$	+2.22	+19.98$_{\pm 0.11}$	+81.07
BERT-base	+0.73$_{\pm 0.15}$	+2.22	+18.94$_{\pm 0.21}$	+40.87

Table 5: AUG minus RAW, in points, with the standard error of the difference as subscript and Cohen's d (Welch, ten random seeds). †: not significant at $p < 0.05$.

Encoder		Identical	Unrelated	Mirrored
DeBERTa-v3-large$_{\text{NLI}}$	RAW	100.00$_{\pm 0.00}$	95.91$_{\pm 0.78}$	75.14$_{\pm 1.20}$
	AUG	100.00$_{\pm 0.00}$	99.73$_{\pm 0.07}$	96.54$_{\pm 0.13}$
DeBERTa-v3-large	RAW	100.00$_{\pm 0.00}$	92.75$_{\pm 0.60}$	73.93$_{\pm 1.46}$
	AUG	99.99$_{\pm 0.04}$	99.71$_{\pm 0.13}$	96.41$_{\pm 1.14}$
BERT-base	RAW	96.46$_{\pm 0.16}$	91.99$_{\pm 0.67}$	49.68$_{\pm 1.74}$
	AUG	100.00$_{\pm 0.00}$	99.33$_{\pm 0.09}$	95.61$_{\pm 0.29}$

Table 6: Accuracy on each sanity suite, in percent, over ten random seeds, for three encoders. Expected class: entailment for identical pairs, neutral for unrelated pairs, contradiction for mirrored pairs.

changes, and whether preservation data could supply the same signal. Table 3 gives macro-F_1, sanity-suite and held-out accuracy per encoder and condition. The best configuration reaches $91.10 \pm 0.14\%$ macro-F_1, 45.46 points above the best baseline, against 16.67% for Majority, 39.36% for Jaccard and 45.64% for TF-IDF (Table 4).

Data augmentation improves the sanity suites, not task performance. Table 5 gives the difference between conditions per encoder with its significance and effect size, and Figure 2 plots both conditions. On the four DeBERTa encoders, the macro-F_1 difference lies between -0.21 and $+0.08$ points and is not significant, and on the other three it is significant but below one point. On the sanity suites, every encoder gains 8.40 to 19.98 points ($p < 10^{-4}$, d from 9.94 to 81.07) and reaches 98.31–98.76%. The suites and the augmented rows come from the same generators, so this shows that the behavior is learnable, not that it generalizes to any degenerate pair.

Per suite (Table 6), identical pairs are solved before augmentation. Unrelated pairs rise from 91.99–95.91% to 99.33–99.73%, and without augmentation the encoders predict 2.38% to 7.87% of them as contradiction, which the signed scale would score negative. Mirrored contradictions rise from 73.93–75.14% to 95.61–96.54%, and from 49.68% on BERT-base.

Paired by random seed, DeBERTa-v3-large$_{\text{NLI}}$ beats its plain counterpart by $+0.19$ points of macro-F_1 ($p = 0.002$, $d = 1.39$) and $+12.14$ points on MoNLI ($p < 10^{-4}$, $d = 2.58$), and at base size the task gain is larger ($+0.67$ points, $p = 0.005$) while MoNLI does not move. MoNLI accuracy stays low: DeBERTa-v3-large$_{\text{NLI}}$ predicts 93.00% of these entailments as neutral and

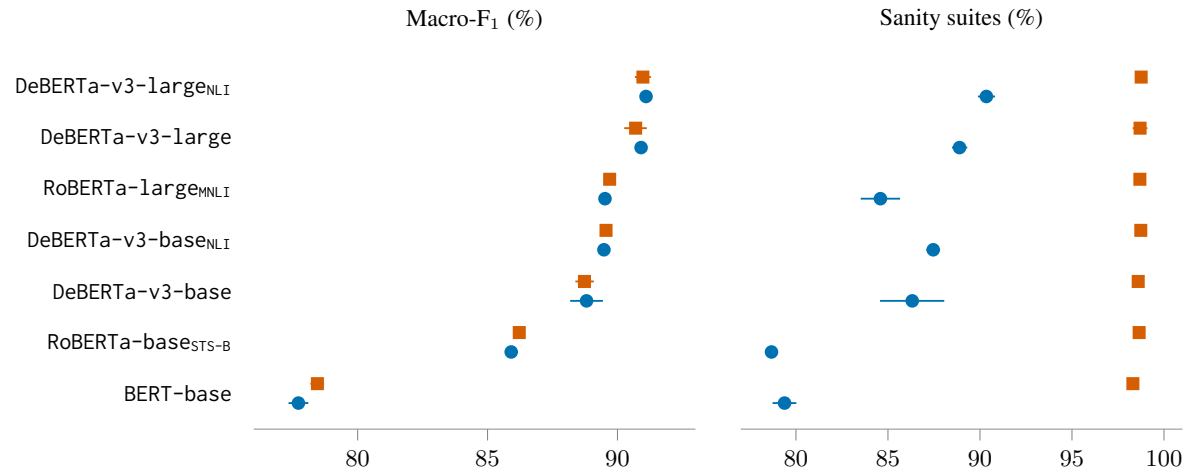


Figure 2: Macro-F_1 (left) and sanity-suite accuracy (right) per encoder. **RAW** and **AUG**: mean over ten random seeds, bars one standard deviation.

0.17% as contradiction, so the heads detect the negation but do not reverse the entailment under it.

The best configuration holds on both corpora: `DeBERTa-v3-large`$_{\texttt{NLI}}$ reaches $91.17 \pm 0.83\%$ macro-F_1 on SICK and $90.35 \pm 0.16\%$ on VitaminC, and $98.11 \pm 1.49\%$ and $98.32 \pm 0.10\%$ AUC, SICK varying more with its 4,116 pairs against 7,884.

Polarity does not follow from preservation data. Table 4 scores three similarity metrics on both tasks, with the preservation column on 1,536 human-rated test pairs of a merged preservation corpus, on whose training split `BERT-base` and `DeBERTa-v3-large` were trained. MeaningBERT ranks the entailment pairs of the test split above its contradiction pairs with an AUC of 60.81%, and of 48.01% on 1,200 public SICK test pairs, where it scores contradictions slightly higher than entailments. `Jaccard` reaches 63.49% with no model at all, so a metric trained on human judgments of meaning preservation ranks agreement above contradiction no better than word overlap. The two columns order the models differently, MeaningBERT ($r = 0.32$) ranking polarity better than `BERT-base` ($r = 0.64$), which sits near chance at 53.73%. A preservation rating has no value for opposite meaning and cannot teach polarity, and `DeBERTa-v3-large` reaches 81.51% through its pretraining.

The off-the-shelf `DeBERTa-v3-large`$_{\texttt{MNLI}}$ already reaches 96.62% AUC. Fine-tuning the same architecture adds +22.74 points of macro-F_1 and +1.78 of AUC, five standard deviations across ten random seeds, so it mostly calibrates the three classes, the ordering being largely available without it, and a practitioner who needs only the sign may skip training.

7 Composing the Two Heads

7.1 The Composed Score on SICK

We compose the best-correlated similarity metric of Table 4, `DeBERTa-v3-large`, with our best polarity head, `DeBERTa-v3-large`$_{\texttt{NLI}}$, by Equation 1, without retraining either. We measure the result on the SICK half of the test split, 4,116 pairs of which 712 are contradictions, and report the mean and standard deviation over the ten heads of Appendix C, using the RAW head as the reference until Section 7.2 introduces the lexical heads. The signed score places $83.88 \pm 2.24\%$ of contradictions below zero, against none for the similarity metric alone, keeps $99.87 \pm 0.07\%$ of entailments and $97.23 \pm 0.17\%$ of neutral pairs above zero, and correlates at $r = 0.78 \pm 0.01$ with human relatedness against 0.85 for the similarity metric alone. SICK neutrals are related captions, and the unrelated case is tested in Table 6. Figure 3 shows the signed score per gold class for one head.

The 16.12% of contradictions that stay positive are not borderline: their median signed score is 48.85 ± 1.80, close to that of neutral pairs, and the head gives them a median p_{contra} of 0.03 ± 0.01. Only $40.71 \pm 6.83\%$ of them contain an explicit negation, against $99.01 \pm 0.21\%$ of the contradictions placed below zero. The others oppose antonyms or contrary events, such as "A little girl is smiling and running outside" and "…running inside", and a few carry debatable labels, where the negated sentence also changes details of the scene. SICK contradictions are thus mostly explicit negations, and the `Negation lookup` places 89.75%

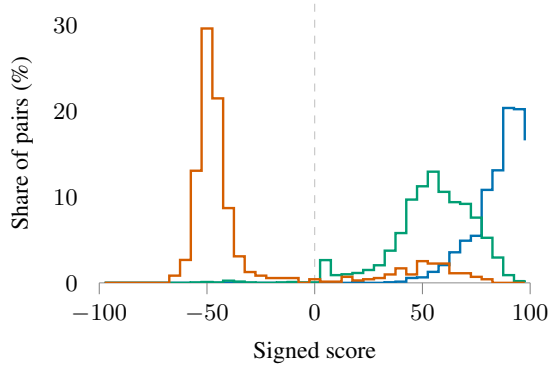


Figure 3: Signed score by gold class (Equation 1, one polarity head), bins of width 5: **entailment**, **neutral**, **contradiction**. Dashed line: zero.

of them below zero, more than the head, but it also sends 16.15% of neutral pairs below zero, and its correlation with relatedness falls to 0.63 and J (Equation 2) to 0.56, against 0.65 ± 0.02 for the head.

7.2 One-Word Edits

To separate negation markers from opposition, we edit one word of each SICK test sentence and check the side of zero the signed score falls on (Table 7). An inserted *not* and an antonym (*outside* for *inside*) should send the pair below zero, and a synonym, a hypernym or a co-hyponym (*cat* for *dog*) should keep it above. The head of Section 6, RAW, catches 97.94% of negations but only 11.22% of unseen antonyms, barely more than the lookup, while the off-the-shelf head catches 87.80% of antonyms but sends 74.70% of co-hyponyms below zero. Our training thus taught negation and lost opposition, which SICK, with only 2,150 of the 152,060 training pairs, barely exercises.

We therefore train a third condition, LEX, which adds to RAW (192,060 pairs in all) 20,000 antonym contradictions and 20,000 synonym or hypernym entailments built by one-word substitution in training sentences. The antonym lexicon is split in two halves, LEX sees one, together with antonyms common in Wikipedia claims such as *increased* and *decreased*, and the other half is held out for Table 7. The entailments keep the head from learning that any changed word is a contradiction. LEX places 73.94% of unseen antonyms below zero, against 11.22%, keeps synonyms and hypernyms above zero and 94.85% of co-hyponyms, and keeps 91.04 ± 0.12% macro-F_1, but its MoNLI accuracy falls from 34.93% to 17.74 ± 3.19%. On SICK, it places 86.14 ± 2.20% of contradictions below zero, for a correlation of 0.74 ± 0.02 and $J = 0.63 \pm 0.01$. Adding the lexical pairs to AUG instead of RAW, AUG+LEX keeps both gains, with 98.96% on the sanity suites and 76.30% of unseen antonyms below zero, for 90.89 ± 0.14% macro-F_1 and 19.20% on MoNLI. On SICK, it places 87.82 ± 0.85% of contradictions below zero, for $r = 0.72 \pm 0.02$, and we release it as the recommended head. When antonym and co-hyponym edits accumulate in the same sentence, its score first drops below zero for 29.10% of single edits and then returns above it, to 0.67% after four, as the pair stops describing the same scene.

7.3 Sensitivity to the Factor

To see what the factor 2 of Equation 1 costs, we replace it with a free factor α, clip the result to $[-100, 100]$, and score each value against three requirements multiplied together, so that sacrificing one cannot hide behind the others:

$$J(\alpha) = c(\alpha) \cdot e(\alpha) \cdot \max(r(\alpha), 0)\,, \qquad (2)$$

with c the share of contradictions below zero, e the share of entailments above it, and r the Pearson correlation with human relatedness on the pairs whose sign does not flip.

A smaller factor gains little (Table 8): $\alpha = 1.5$ scores 2.00% higher on J but truncates the negative half at -50, so a contradiction can never score as far below zero as its paraphrase scores above it. A larger factor gains little either, $\alpha = 3$ placing 1.24 more points of contradictions below zero for 0.06 of correlation. With one head, on the development split, and MeaningBERT in place of `DeBERTa-v3-large`, J again peaks at $\alpha = 1.5$ and, among the factors that reach -100, at $\alpha = 2$. Since the sign depends on p_{contra} alone, only r depends on the similarity metric.

7.4 What Fine-Tuning the Polarity Head Buys

With the off-the-shelf head (Appendix D), 97.61% of contradictions fall below zero against 83.88%, but the correlation with relatedness drops by 0.18 and J by 12.05% relative. Figure 4 shows why: among the pairs it gives a contradiction probability near 0.5, fewer than one in ten is a contradiction, so the product lowers the score of merely unrelated pairs.

7.5 What the Sign Buys at the Decision Point

Table 9 gives, at three acceptance thresholds, the share of accepted pairs that are contradictions and

Edit	Negation	Antonym, seen	Antonym, unseen	Synonym	Hypernym	Co-hyponym
Expected sign	< 0	< 0	< 0	> 0	> 0	> 0
Pairs	2000	1560	254	2000	2000	2000
Similarity alone	0.00	0.00	0.00	100.00	100.00	100.00
`Negation lookup`	99.85	0.00	0.00	100.00	100.00	100.00
Off-the-shelf head	97.00	88.08	87.80	100.00	99.95	25.30
Ours, RAW	$97.94_{\pm 0.72}$	$4.86_{\pm 1.67}$	$11.22_{\pm 2.72}$	$99.99_{\pm 0.03}$	$99.97_{\pm 0.06}$	$99.73_{\pm 0.12}$
Ours, LEX	$99.50_{\pm 0.62}$	$87.15_{\pm 3.56}$	$73.94_{\pm 5.43}$	$100.00_{\pm 0.00}$	$100.00_{\pm 0.02}$	$94.85_{\pm 1.32}$
Ours, AUG+LEX	$99.67_{\pm 0.25}$	$87.51_{\pm 3.10}$	$76.30_{\pm 3.38}$	$100.00_{\pm 0.00}$	$100.00_{\pm 0.00}$	$95.84_{\pm 1.63}$

Table 7: One-word edits of SICK test sentences: percent of pairs on the expected side of zero. Ours: mean with standard deviation as subscript over ten heads. Unseen antonyms are held out of LEX.

Factor	Contra. < 0	Neutral > 0	Rel. r	Floor
$\alpha = 1.00$	$0.00_{\pm 0.00}$	$98.35_{\pm 0.00}$	$0.83_{\pm 0.00}$	0
$\alpha = 1.25$	$81.21_{\pm 3.21}$	$97.73_{\pm 0.09}$	$0.82_{\pm 0.00}$	-25
$\alpha = 1.50$	$82.74_{\pm 2.72}$	$97.49_{\pm 0.12}$	$0.81_{\pm 0.01}$	-50
$\alpha = 1.75$	$83.36_{\pm 2.43}$	$97.37_{\pm 0.13}$	$0.79_{\pm 0.01}$	-75
$\boldsymbol{\alpha = 2.00}$	$\mathbf{83.88}_{\pm 2.24}$	$\mathbf{97.23}_{\pm 0.17}$	$\mathbf{0.78}_{\pm 0.01}$	$\mathbf{-100}$
$\alpha = 3.00$	$85.11_{\pm 1.84}$	$96.70_{\pm 0.22}$	$0.72_{\pm 0.01}$	-100
Magnitude alone	0.00	98.35	0.85	0

Table 8: Composed score per factor α in place of the 2 of Equation 1, SICK test half (1,404 entailments, 2,000 neutral, 712 contradictions), in percent, mean with standard deviation as subscript over ten polarity heads. **Bold**: the factor of Equation 1.

Scale	$s > 25$	$s > 50$	$s > 70$
Contradictions accepted			
Magnitude alone	17.83	14.10	0.91
Negation lookup	2.36	2.30	0.85
Off-the-shelf head	0.49	0.15	0.00
Ours, RAW	$2.99_{\pm 0.29}$	$2.03_{\pm 0.09}$	$0.47_{\pm 0.07}$
Ours, LEX	$2.55_{\pm 0.32}$	$1.42_{\pm 0.18}$	$0.13_{\pm 0.04}$
Ours, AUG+LEX	$2.29_{\pm 0.16}$	$1.26_{\pm 0.11}$	$0.10_{\pm 0.03}$
Entailments kept			
Magnitude alone	100.00	99.15	88.32
Negation lookup	99.64	99.00	88.32
Off-the-shelf head	97.86	96.72	86.18
Ours, RAW	$99.74_{\pm 0.11}$	$98.17_{\pm 0.16}$	$86.28_{\pm 0.56}$
Ours, LEX	$99.73_{\pm 0.11}$	$98.31_{\pm 0.13}$	$86.93_{\pm 0.40}$
Ours, AUG+LEX	$99.80_{\pm 0.10}$	$98.48_{\pm 0.19}$	$87.23_{\pm 0.12}$

Table 9: Contradictions accepted and entailments kept by each scale at three thresholds, in percent. Percentile bootstrap intervals (1,000 resamplings) are at most ± 1.2 wide; ours: mean with standard deviation as subscript over ten heads.

the share of entailments kept. At a threshold of 50, 14.10% of the pairs the similarity metric accepts are contradictions, against $2.03 \pm 0.09\%$ for RAW, $1.42 \pm 0.18\%$ for LEX and $1.26 \pm 0.11\%$ for AUG+LEX, a sevenfold to elevenfold reduction with non-overlapping bootstrap intervals, and 2.30% for the lookup. Entailment recall falls only from 99.15% to between 98.17% and 98.31%, so the signed scale does not simply lower every score. Across thresholds from 0 to 95 in steps of five, RAW never accepts more contradictions than the similarity metric and never loses more than 3.85 points of recall on average. The off-the-shelf head accepts 0.15% of contradictions for 96.72% of entailments, so the gain comes from the shape of the scale and fine-tuning buys recall. On simplification output, where contradictions are rare, the sign selects the pairs humans rated worst (Appendix E).

8 Conclusion

Meaning preservation metrics output a non-negative score, so they cannot separate a contradiction from a pair that shares little meaning, and a published metric ranks entailment above contradiction no better than word overlap. We proposed a signed scale on $[-100, 100]$ that keeps any similarity metric as it is and multiplies its score by $1 - 2\,p_{\text{contra}}$, so the closeness of a pair sets the absolute value and a polarity head sets the sign. The polarity head is learnable on a stratified, leakage-controlled inference corpus, reaching 91.10% macro-F_1, and generated identical, unrelated and mirrored pairs make it reliable at both ends of the scale. Composed with an existing metric, the signed score places 83.88% of SICK contradictions below zero, where the similarity metric alone places none, and accepts seven times fewer contradictions at a fixed threshold while keeping almost every entailment. One-word edits show what a word list cannot do: a head trained with generated antonym contradictions places 73.94% of unseen antonyms below zero, against 11.22% without them and none for a negation lookup. Combined with the generated identical and unrelated pairs, it keeps the ends of the scale and cuts accepted contradictions elevenfold, at a cost of 0.21 points of macro-F_1 and of correlation with relatedness.

Limitations

Seven limitations bound our claims.

First, there is no signed gold standard. The composition is evaluated against three necessary conditions rather than against human signed judg-

ments, because none exist. A corpus of human-rated signed pairs would let the form of Equation 1 be tested rather than argued, and collecting one is the natural next step.

Second, data augmentation costs monotonicity. The gain in sanity is paid for on MoNLI, which drops on three of seven encoders and on both large DeBERTa variants, by -6.82 points on the plain one and -3.69 on the inference-pretrained one. Generated identical and unrelated pairs teach the easy ends of the scale, not the reversal of entailment under negation, and the lexical conditions halve MoNLI accuracy again, from 34.93% to 17.74% and 19.20%.

Third, a single corpus carries the composition. Every measurement of the signed scale lives on the SICK half of the test split, 4,116 pairs of which 712 are contradictions, which is also where the sensitivity to the factor is measured. The constraint is structural rather than a choice: SICK is the only corpus we know of annotating relatedness and inference over the same pairs, and the third requirement of Equation 2 needs both. It binds the sensitivity analysis unavoidably and the evaluation only by convenience, since Table 9 needs inference labels alone and could in principle be measured elsewhere. The split is also thin on the development side: SICK supplies 495 development rows, of which 229 shared source sentences with its own test portion and were removed, leaving the development split at 2.27% SICK against 34.30% in test. Model selection is therefore driven by VitaminC while the composition is analyzed on SICK alone, and the two numbers are the first thing to watch should they diverge.

Fourth, the sign costs correlation. On both human-rated quantities the composition correlates worse than the magnitude alone: 0.78 against 0.85 on SICK relatedness, and 0.64 against 0.70 on preservation ratings. This is what should happen, since neither rating scale is signed and a correct negative score can only move a pair away from a non-negative target. Correlation with a non-negative rating is therefore the wrong instrument here, and Table 9 measures the benefit where it exists instead. What no existing human judgement can settle is whether the magnitude of a negative score is right, which a signed annotation would. That magnitude also depends on where m comes from: a preservation metric already lowers m for the negation it partly detects, so with MeaningBERT the negation of Figure 1 would land at -58.30 rather than -100, whereas a relatedness metric would not count the negation twice.

Fifth, the sign rests on one head. Because the composition is a product with $m \geq 0$, the sign of the score is decided entirely by p_{contra}, and a related pair on which the head is undecided, at $p_{\text{contra}} = 0.5$, scores 0 like an unrelated one. A polarity head that is wrong about a pair is not corrected by a good similarity metric, and the 16.12% of SICK contradictions that stay positive are exactly the pairs the head misses, mostly contradictions without an explicit negation (Section 7). The one-word edits that measure this rest on hand-written lexicons for SICK's vocabulary, and a few antonym swaps, such as *large* and *small* gathering, are contrary only under the same-scene convention. The lexical conditions learn antonyms from that same convention, and 23.70% to 26.06% of unseen antonyms still stay positive. We also designed them after seeing the RAW head fail on these edits, so the held-out half of the lexicon guards against memorized word pairs but not against a test built with the remedy in mind.

Sixth, the property the title names is never measured on natural unrelated pairs. That two sentences about different topics score near zero follows from the product form, since $m \approx 0$ leaves the polarity term nothing to act on, and it is measured on the generated unrelated suite. No corpus we know of annotates genuinely unrelated pairs for meaning preservation, because such a pair is not an interesting annotation, so the construction carries this half of the claim and the evidence for it is self-generated.

Finally, the work is English and sentence-level throughout. Every corpus we merge is English, and the scale is defined over sentence pairs. Neither restriction is incidental. The polarity half rests on inference corpora, which are abundant in English and thin in most other languages, so the composition cannot be ported by translating a metric alone. Contradiction between documents is also a different relation from contradiction between sentences, since a document can deny another in one passage and agree with it everywhere else, and a single scalar cannot carry that. Whether the product form survives either move is untested.

Ethical Considerations

A meaning preservation metric is deployed to decide whether a rewritten sentence may be shown

to a reader in place of the original. When such a metric reports that a negated sentence preserves the meaning of its source, as we measure in Section 6, the error reaches the reader rather than the engineer. The risk is concrete in the settings where simplification is most useful, where a layperson may rely on the rewritten text without access to the original (Beauchemin et al., 2023). A signed scale does not remove that risk, but it makes the failure visible to the system consuming the score instead of hiding it inside a positive number. Our corpora also inherit the biases of the Wikipedia revisions and caption corpora they are drawn from, and a polarity head trained on them carries those biases into whatever it gates (Bender et al., 2021).

Availability

The corpus builder, the loaders, the training and composition code, the result files of every cell and the script generating every table and figure in this paper will be released under an open licence. Repository: removed for double-blind review. The merged corpus inherits the most restrictive licence among its sources and will be redistributed under CC BY-NC-SA 4.0. The loaders are released separately so the corpus can be rebuilt from the original sources.

Use of Large Language Models

This work used Large Language Models (specifically Claude) to assist with the linguistic refinement, typographical errors, and LaTeX formatting of the manuscript (table generation, figure improvements). The authors retained full control over the scientific content, data analysis, and conclusions.

Acknowledgements

Removed for double-blind review.

References

Gaurab Baral. 2026. LexFlip: A Dissociation Diagnostic for Legal Meaning Preservation Metrics. *arXiv preprint arXiv:2609.05296*.

David Beauchemin, Michelle Albert-Rochette, Richard Khoury, and Pierre-Luc Déziel. 2025. JUDGEBERT: Assessing legal meaning preservation between sentences. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing*, pages 92–118.

David Beauchemin, Horacio Saggion, and Richard Khoury. 2023. MeaningBERT: Assessing Meaning Preservation Between Sentences. *Frontiers in Artificial Intelligence*, 6.

Emily M. Bender, Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell. 2021. On the Dangers of Stochastic Parrots: Can Language Models Be Too Big? In *Proceedings of the ACM Conference on Fairness, Accountability, and Transparency*, pages 610–623.

Yanran Chen and Steffen Eger. 2023. MENLI: Robust evaluation metrics from natural language inference. *Transactions of the Association for Computational Linguistics*, 11:804–825.

Marie-Catherine de Marneffe, Anna N. Rafferty, and Christopher D. Manning. 2008. Finding contradictions in text. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics*, pages 1039–1047.

Ashwin Devaraj, William Sheffield, Byron Wallace, and Junyi Jessy Li. 2022. Evaluating factuality in text simplification. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics*, pages 7331–7345.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In *Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics*, pages 4171–4186.

Atticus Geiger, Kyle Richardson, and Christopher Potts. 2020. Neural Natural Language Inference Models Partially Embed Theories of Lexical Entailment and Negation. In *Proceedings of the BlackboxNLP Workshop on Analyzing and Interpreting Neural Networks for NLP*, pages 163–173.

James A. Hanley and Barbara J. McNeil. 1982. The Meaning and Use of the Area under a Receiver Operating Characteristic (ROC) Curve. *Radiology*, 143(1):29–36.

Pengcheng He, Jianfeng Gao, and Weizhu Chen. 2023. DeBERTaV3: Improving DeBERTa using ELECTRA-Style Pre-Training with Gradient-Disentangled Embedding Sharing. In *International Conference on Learning Representations*.

Felix Hill, Roi Reichart, and Anna Korhonen. 2015. SimLex-999: Evaluating semantic models with (genuine) similarity estimation. *Computational Linguistics*, 41(4):665–695.

Sture Holm. 1979. A Simple Sequentially Rejective Multiple Test Procedure. *Scandinavian Journal of Statistics*, 6(2):65–70.

Paul Jaccard. 1912. The Distribution of the Flora in the Alpine Zone. *New Phytologist*, 11(2):37–50.

Philippe Laban, Tobias Schnabel, Paul N. Bennett, and Marti A. Hearst. 2022. SummaC: Re-visiting NLI-based models for inconsistency detection in summarization. *Transactions of the Association for Computational Linguistics*, 10:163–177.

Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. 2019. RoBERTa: A Robustly Optimized BERT Pretraining Approach. *arXiv preprint arXiv:1907.11692*.

Chen Lyu and Gabriele Pergola. 2024. SciGisPy: A novel metric for biomedical text simplification via gist inference score. In *Proceedings of the Workshop on Text Simplification, Accessibility and Readability*, pages 95–106.

Mounica Maddela, Yao Dou, David Heineman, and Wei Xu. 2023. LENS: A Learnable Evaluation Metric for Text Simplification. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics*, pages 16383–16408.

Henry B. Mann and Donald R. Whitney. 1947. On a Test of Whether One of Two Random Variables is Stochastically Larger than the Other. *The Annals of Mathematical Statistics*, 18(1):50–60.

Marco Marelli, Stefano Menini, Marco Baroni, Luisa Bentivogli, Raffaella Bernardi, and Roberto Zamparelli. 2014. A SICK Cure for the Evaluation of Compositional Distributional Semantic Models. In *Proceedings of the International Conference on Language Resources and Evaluation*, pages 216–223.

Joshua Maynez, Shashi Narayan, Bernd Bohnet, and Ryan McDonald. 2020. On faithfulness and factuality in abstractive summarization. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics*, pages 1906–1919.

Saif M. Mohammad, Bonnie J. Dorr, Graeme Hirst, and Peter D. Turney. 2013. Computing lexical contrast. *Computational Linguistics*, 39(3):555–590.

Yixin Nie, Adina Williams, Emily Dinan, Mohit Bansal, Jason Weston, and Douwe Kiela. 2020. Adversarial NLI: A New Benchmark for Natural Language Understanding. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics*, pages 4885–4901.

Nils Reimers and Iryna Gurevych. 2019. Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing*, pages 3980–3990.

Horacio Saggion. 2017. *Automatic Text Simplification*. Morgan & Claypool.

Tal Schuster, Adam Fisch, and Regina Barzilay. 2021. Get Your Vitamin C! Robust Fact Verification with Contrastive Evidence. In *Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics*, pages 624–643.

Thomas Scialom, Paul-Alexis Dray, Patrick Gallinari, Sylvain Lamprier, Benjamin Piwowarski, Jacopo Staiano, and Alex Wang. 2021. QuestEval: Summarization Asks for Fact-based Evaluation. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing*, pages 6594–6604.

Marina Sokolova and Guy Lapalme. 2009. A Systematic Analysis of Performance Measures for Classification Tasks. *Information Processing & Management*, 45(4):427–437.

James Thorne, Andreas Vlachos, Christos Christodoulopoulos, and Arpit Mittal. 2018. FEVER: A Large-scale Dataset for Fact Extraction and VERification. In *Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics*, pages 809–819.

Thinh Hung Truong, Yulia Otmakhova, Timothy Baldwin, Trevor Cohn, Jey Han Lau, and Karin Verspoor. 2022. Not Another Negation Benchmark: The NaN-NLI Test Suite for Sub-clausal Negation. In *Proceedings of the Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics*, pages 883–894.

B. L. Welch. 1947. The Generalization of Student's Problem when Several Different Population Variances are Involved. *Biometrika*, 34(1-2):28–35.

Adina Williams, Nikita Nangia, and Samuel Bowman. 2018. A Broad-Coverage Challenge Corpus for Sentence Understanding through Inference. In *Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics*, pages 1112–1122.

Yuheng Zha, Yichi Yang, Ruichen Li, and Zhiting Hu. 2023. AlignScore: Evaluating factual consistency with a unified alignment function. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics*, pages 11328–11348.

Tianyi Zhang, Varsha Kishore, Felix Wu, Kilian Q. Weinberger, and Yoav Artzi. 2020. BERTScore: Evaluating Text Generation with BERT. In *International Conference on Learning Representations*.

A Corpus Construction and Leakage Control

A.1 Merging and Harmonization

The two training corpora, VitaminC and SICK, are merged into one table with a unified three-way polarity column. MoNLI and NaN-NLI are kept apart and never enter it. VitaminC carries FACT3 labels, which we map onto NLI3 as SUPPORTS to entailment, NOT ENOUGH INFO to neutral and REFUTES to contradiction. SICK is assembled from two public halves, one carrying the inference label and one the relatedness score. The join is verified to lose no more than 1% of rows, and in practice loses none.

A.2 Stratification

We keep the train, development and test splits that each corpus publishes. The development and test splits are then reduced to 4,000 rows per class, drawn in equal share from each corpus by water-filling: a corpus smaller than its share is taken whole and the shortfall is redistributed over the larger ones. Without it, SICK falls to 8.16% of the test split and 0.78% of the development split.

A.3 Leakage

VitaminC reuses one piece of evidence across many claims, and its published splits do not separate evidence sentences, so 1,662 source sentences, 5.6% of the test groups, were also present in training. Once the stratified test split is fixed, we remove the rows that share a source sentence with a later split: 229 development rows whose source sentence appears in test, then 2,379 training rows whose source sentence appears in development or test. Rows are removed from development and training rather than from test so that the test split stays fixed and balanced. Development only selects models and can afford the loss. We then assert that train/dev, train/test and dev/test share no pair and no source sentence, and run the assertion again after data augmentation, since a generated pair inherits its source sentences.

A.4 Data Augmentation Quotas

Each generated kind of Section 4 feeds one class: identical pairs feed entailment, unrelated pairs feed neutral and mirrored contradictions feed contradiction. Each contributes exactly 25,000 pairs, so every class grows by the same amount and the training split stays balanced, going from 152,060 to 227,060 rows. The quota is needed because the three kinds are not naturally equal in number: mirroring every contradiction would add 50,267 pairs to that class alone, doubling it to 100,534 against 50,602 entailments. To reach the quota, each generator is asked for 40% more pairs than needed, since the unrelated generator rejects draws whose sentences overlap too much and the identical one skips sentences already used, and the surplus is discarded at random.

B Measuring Polarity in a Similarity Metric

The similarity-metric measurements of Section 6 train nothing and are deterministic, so they involve no random seed.

B.1 Model

We score MeaningBERT (Beauchemin et al., 2023) with its public checkpoint, without fine-tuning or recalibration. For each pair, it returns one meaning preservation score on $[0, 100]$, and that score is the only quantity used.

B.2 Data

The main measurement uses the test split of Table 2: 12,000 pairs, 4,000 per class, drawn from SICK and VitaminC by the stratification of Appendix A. The AUC uses its 4,000 entailment and 4,000 contradiction pairs. Neutral pairs are excluded, since a similarity metric is expected to score them low and they say nothing about polarity. The SICK figure uses a separate sample: the first 600 entailment and the first 600 contradiction pairs of the public SICK test set, in their published order. On this sample the checkpoint scores entailments at 67.34 on average and contradictions at 69.18.

B.3 Statistic

The AUC is the probability that a randomly drawn entailment pair receives a higher score than a randomly drawn contradiction pair, counting a tie as one half. We compute it from mid-ranks through its equivalence with the Mann-Whitney statistic (Mann and Whitney, 1947; Hanley and McNeil, 1982). Ties matter here because a clamped output head saturates at exactly 0 and 100, and ordinal ranks would break those ties by position in the array. An AUC of 50% means the score does not order the two classes, and an AUC below 50% means it orders them backwards.

B.4 Surface Baseline

Both sentences are lowercased and split into alphanumeric tokens, and the score of a pair is the Jaccard overlap of the two token sets. This score enters the AUC directly and involves no training. The two thresholds of Section 6, fitted on the development split, serve only to turn the same score into three classes for macro-F_1.

C Training Details and Compute Budget

C.1 Optimization

Table 10 gives the settings, which are identical for all 140 runs. The library version is part of them: a later major release of `transformers` sends

`DeBERTaV3-large` to a non-finite gradient within a hundred steps, on the same corpus and the same settings, and reports the majority-class floor rather than failing. Nothing is tuned per encoder: one setting across the grid is what makes the seven encoders comparable, and the cost is that no encoder sits at its own optimum.

Setting	Value
Objective	cross-entropy, three classes
Optimizer	AdamW, default β
Learning rate	10^{-5}, linear decay
Epochs	2
Effective batch	32
Micro-batch	8 (base), 4 (large)
Gradient accumulation	4 (base), 8 (large)
Max sequence length	256 tokens
Precision	fp32
Library	`transformers` 4.57
Random seeds	$[42, \ldots, 51]$

Table 10: Optimization settings, identical across the grid.

C.2 Head Reuse

Four of the seven checkpoints ship a three-way inference head. We keep it when its class order matches ours and permute its rows when the published order is reversed, as it is for `roberta-large-mnli`. The alternative, training from a permuted head, is silent and costs an epoch of relearning the permutation.

C.3 Random Seeds

Random seeds $[42, \ldots, 51]$ control initialization of the new head, data order and dropout. Unless stated otherwise, every number reported in this paper is a mean over the ten with its standard deviation. The AUC column of Table 4 and the per-corpus figures of Section 6 come from ten further runs of the best cell that keep their weights, since the grid stores predicted classes and an AUC needs probabilities. Nine of those ten use a micro-batch of 8 with accumulation 4 on a later card, the effective batch unchanged at 32. Section 7 composes the same ten heads. Figure 3, Figure 4, the comparison with MeaningBERT and the check on simplification output use the head of random seed 42 alone.

C.4 Hardware

The grid runs on three NVIDIA Pascal-generation cards: a GTX 1080 Ti (11 GB), a Quadro P5000 (16 GB) and a Titan Xp (12 GB). All training is fp32, since the generation has no usable bfloat16. The ten LEX heads of Section 7 and the ten AUG+LEX heads were trained later on three NVIDIA RTX 6000 Ada cards, in fp32 with a micro-batch of 16 and accumulation 2, the effective batch unchanged at 32, each run taking about 85 and 115 minutes.

Polarity head	Off-the-shelf	Fine-tuned
Contradictions below zero	**97.61**	$83.88_{\pm 2.24}$
Relatedness r	0.60	$\mathbf{0.78}_{\pm 0.01}$
J (Equation 2)	0.57	$\mathbf{0.65}_{\pm 0.02}$

Table 11: One similarity metric, two polarity heads, SICK test half. Fine-tuned: mean with standard deviation as subscript over ten heads. First row in percent of the 712 contradictions. Higher is better.

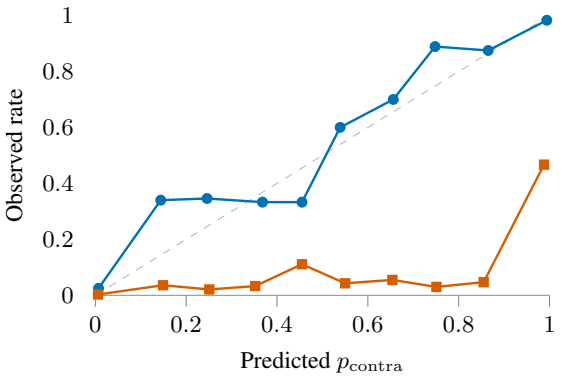


Figure 4: Observed contradiction rate against predicted p_{contra}, SICK test half, ten bins, for the **fine-tuned head** and the **off-the-shelf head**. Dashed diagonal: perfect calibration.

C.5 Budget

The grid takes roughly 500 GPU-hours in total. A base-encoder run takes 68 minutes on the RAW condition and 109 minutes on AUG. A large-encoder run takes 318 and 489 minutes. The composition of Section 7 trains nothing: it is two forward passes over 4,116 pairs plus the sensitivity sweep of Section 7, and runs on CPU in under half an hour. No private API was used, and the total monetary cost is the electricity of hardware we already own.

D Composition with an Off-the-Shelf Head

The two heads are composed with the same similarity metric by Equation 1 and scored on the SICK test half. Table 11 presents the trade-off between the sign criterion, relatedness and the combined objective for the two heads, and Figure 4 presents the reliability of p_{contra} for each. The off-the-shelf head wins the sign criterion and loses the combined objective, because it is confident on pairs that are merely unrelated: in the bins where it predicts a contradiction probability between 0.4 and 0.9, the

observed rate stays near one in ten, so the product subtracts magnitude from pairs that should keep theirs. The fine-tuned head tracks the diagonal over the same range, which is what the third requirement of Equation 2 rewards.

E The Scale on Simplification Output

On the 407 human-annotated test pairs of CSMD, the signed scale with one head sends 4.18% below zero, with a mean human rating of 38.29 against 70.97 for the rest: the sign selects the simplifications humans rated worst. On this corpus without deliberate contradictions, it costs 0.06 of correlation with the human rating, 0.64 against 0.70 for the similarity metric alone.